BioRaman

Raman Hyperspectral Map Analysis for Biophysics — User Manual

Version 0.8.0 • Application file: `bioraman.py`
Audience: new and experienced users • Desktop GUI (Python / Tkinter)

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A desktop application for loading, visualising and analysing Raman hyperspectral maps — with preprocessing, univariate and ratio mapping, curve fitting, PCA, clustering, MCR-ALS, N-FINDR endmember extraction, peak identification and 3D confocal rendering.

Contents

1 Introduction

BioRaman is a desktop application for working with Raman hyperspectral maps — datasets where a full Raman spectrum is recorded at every pixel of a 2D (or 3D) sample area. It reads Renishaw WiRE `.wdf` files and tabular `.xlsx` data, and provides a complete workflow from raw spectra to publication-ready chemical maps and statistics.

The program is organised around a single **main window** showing the map on one side and the spectrum at the cursor on the other. All advanced analyses (PCA, clustering, unmixing, 3D rendering, and so on) open in their own dedicated windows from the menu bar, so you can keep several analyses open at once.

How to use this manual

Sections 2–9 cover the everyday workflow and are recommended reading for **new users**. Sections 10–17 describe the advanced analysis windows and are written for users already familiar with chemometrics and spectroscopy, though each begins with a plain-language summary. Sections 18–21 are reference material.

Note BioRaman follows semantic versioning. Any change that affects analytical output is flagged in the project CHANGELOG and triggers a new version number, so results can always be traced to a specific release.

2 Installation & Launching

Requirements

Python 3.9 or newer is required. The application depends on a small set of scientific Python packages.

Install the dependencies

From a terminal, install the required packages with pip:

```
pip install numpy scipy matplotlib pybaselines renishawWiRE pillow
pip install scikit-learn pandas seaborn openpyxl
```

The 3D viewer uses `mpl_toolkits.mplot3d`, which ships with matplotlib and needs no separate installation. Two optional libraries enhance functionality if present: **pybaselines** (advanced baseline correction) and **renishawWiRE** (reading `.wdf` files). The program detects these automatically and disables the related controls if they are missing.

Launch the application

Run the program from the folder that contains it:

```
python bioraman.py
```

3 The Main Window at a Glance

When the application starts you see the main Raman Explorer window, divided into four zones:

Zone	What it contains
Menu bar	**File, View, Preprocessing, Analysis, Help** menus — the entry point to every feature and analysis window.
Toolbar	Quick-access buttons (Open WDF, Save Map, Save Spectrum, and the analysis launchers) plus the **MODE** selector, the **CMAP** (colour-map) selector and a progress bar.
Map panel (left)	The colour-coded chemical map of the sample. You click here to pick a pixel, and draw ROIs here.

Zone	What it contains
Spectrum panel (right)	The Raman spectrum at the selected pixel (or the mean spectrum of a drawn ROI). Right-click adds a spectrum to the comparison overlay.

The MODE selector

The **MODE** radio buttons control what the main map shows: **Ratio** (a band-ratio or single-band intensity map), **A+B RGB** (two bands rendered as colour channels) and **White Light** (the optical reference image, if loaded).

The CMAP selector

Choose the colour map (look-up table) used to render intensities — `turbo` by default. This is purely a display choice and does not alter the underlying data.

4 Loading Data

Open a Raman map

Choose **File › Open WDF…** (or press **Ctrl+O**) and select a Renishaw `.wdf` map file. The map renders in the left panel and the spectrum at the first pixel appears on the right. Wavenumbers are shown in cm^{-1}.

Load a white-light (optical) image

Choose **File › Load White-Light…** to overlay or compare an optical micrograph of the same area. Switch the MODE selector to **White Light** to view it.

Note Large data files (raw spectra, exports) are intentionally excluded from the code repository. Keep your `.wdf` data in a separate data folder.

5 Preprocessing

Preprocessing cleans the raw spectra before any mapping or analysis. Open **Preprocessing › Settings…** to configure the pipeline. Settings apply to the **next** file you load, so set them before opening data (or reload after changing).

Step	Option	What it does
Cosmic-ray removal	Enable	Removes sharp spurious spikes using a modified Z-score on the first derivative (Whitaker & Hayes, 2018).
Dark / offset removal	Enable	Subtracts the detector offset and stray-light pedestal.
Baseline correction	asls / arpls / drpls / none	Removes fluorescence and broad background. *asls* is a good general choice; *arpls* suits strong or sloping backgrounds; *drpls* suits broad peaks.
Smoothing	Enable	Savitzky–Golay smoothing to reduce noise.
Normalisation	max / area / none	*max* scales to the tallest peak; *area* scales to total integrated intensity; *none* keeps raw counts.

Click **Apply & Close** to save the settings, or **Cancel** to discard. To see exactly what was applied to the loaded data, open **Preprocessing › Processing Log…**.

Note Preprocessing choices change analytical output. Record the settings you used (the Processing Log lists them) so your maps and statistics are reproducible.

6 Viewing & Navigating Spectra

- **Pick a pixel:** click anywhere on the map; its spectrum appears on the right.
- **Compare spectra:** right-click the map to add the clicked spectrum to a comparison overlay. Use **Clear comparison** to reset.
- **Peak markers:** toggle **View › Show peak markers** to annotate detected peaks.
- **Normalise view:** toggle **View › Normalise spectrum** to compare peak shapes independent of absolute intensity.
- **Save the current spectrum: File › Save Spectrum** (Ctrl+S).

7 Building Maps

Maps turn the spectral data cube into a single colour image where each pixel's colour encodes a chemical quantity. Several map builders are available.

7.1 Univariate Analysis

Open from **Analysis** (Univariate). Build a map from a single spectral feature.

Control	Meaning
Map type	Quantity to map — e.g. peak intensity, integrated band area, or intensity at a single point.
First limit (cm^{-1})	The band position, or the lower edge of an integration window.
Second limit (cm^{-1})	The upper edge of the band window. (Ignored for ‘Intensity at Point’, which uses only the first limit.)
Save map as	Name under which the generated map is stored for later use (e.g. as an input to a ratio map).

Click **Create Map** to generate it.

7.2 Ratio Map

Open the Ratio Map window to divide one stored map by another — a common way to normalise out thickness or focus variation. Choose a **Numerator map** and a **Denominator map** from maps you have already created, name the result, and click **Create Ratio Map**.

7.3 Dynamic Mapping

The Dynamic Mapping window builds maps interactively across acquisitions, with live ROI tools, an adjustable white-light overlay (with opacity control), and options to save the ROI image or ROI mask directly.

7.4 Curve-Fit Map

The Curve-Fit Map window fits a peak model to every pixel and maps a fitted parameter (such as position, width or amplitude). Set minimum peak width and height limits to avoid degenerate fits, then click **Run Fit & Create Map**. Fitting every pixel is computationally heavy and may take time on large maps.

8 Regions of Interest (ROI) & ROI Analysis

An ROI restricts analysis to a chosen area of the sample. On the main map use **Draw ROI**, choose a shape (**rectangle, ellipse, polygon** or **freehand**), and draw on the map. **Clear** removes it. Tick **Mask outside ROI** to hide everything beyond the region.

Click **Analyse ROI** to open the **ROI Analysis** window, a guided spectroscopic workflow: *define bands* → *binarize* → *heatmaps* → *statistics*.

- **Define bands:** set the spectral bands (A, B, …) that characterise your components.
- **Auto-set threshold:** a one-click threshold (30th percentile of Band A) to separate signal from background.
- **Rubber-band baseline:** recommended for weak bands.
- **Run Analysis:** generates binarized maps, heatmaps and per-region statistics.
- **Export Figure / Export Panels:** save the composite figure or the individual panels.

9 Colour & Display Control (LUT)

The LUT (look-up table) Control window fine-tunes how intensities map to colour, without changing the data.

- **Colour scheme:** choose the colour map.
- **Opacity:** blend the map over the white-light image.
- **vmin / vmax:** set the lower and upper intensity limits manually.
- **Auto** and **5%–95%**: auto-scale the limits, optionally clipping outliers to the 5th–95th percentile for better contrast.

10 PCA Analysis

In brief: Principal Component Analysis (PCA) re-expresses many correlated spectra in terms of a few uncorrelated **principal components** (PCs) — the directions along which the spectra vary most. The first component captures the largest share of variation, the second the next largest (independent of the first), and so on. PCA is the standard first look at a dataset: it reveals whether spectra fall into distinct groups, highlights outliers, and shows which spectral regions drive the differences — all without needing reference standards.

Open from **Analysis › PCA Analysis…**. The window has a control panel on the left and a multi-panel figure on the right.

Step 1 — Add input data

Unlike the other analysis windows, PCA works across one or more files so you can compare samples. Add data with **+ Add WDF** (a Renishaw map) or **+ Add XLSX** (a tabular spectra file); select an entry and press **Remove** to drop it. Each file becomes a labelled group in the score plots.

Step 2 — Label your groups

Select a file in the list, type a name in the **Label selected** box and click **Set**. These labels are what appear in the legend of the score plot, so give them meaningful names (e.g. ‘control’, ‘treated’).

Step 3 — Set the options

Control	Default	Meaning
Components	3	Number of principal components to compute (2–10).
Wavenumber min	500	Lower edge of the spectral window used (cm^{-1}).

Control	Default	Meaning
Wavenumber max	3500	Upper edge of the spectral window used (cm^{-1}).
Spectra/file (0 = all)	0	Randomly sub-sample this many spectra per file to speed up very large datasets; 0 uses every spectrum.
Remove outliers (>2σ)	On	Discard spectra lying beyond two standard deviations before fitting, so extreme pixels do not dominate.
Standardise features	Off	Scale every wavenumber channel to unit variance before PCA. Use this when you care about peak *shape* rather than absolute intensity; leave off to let strong bands carry more weight.

Step 4 — Run and read the results

Click **Run PCA**. The figure on the right shows five panels:

Panel	What it shows	How to read it
A — Scores (PC1 vs PC2)	Each spectrum as a point, coloured by group label. Axis labels include the % variance each PC explains.	Points that cluster together are spectrally similar; separated clouds indicate distinct groups.
B — Loadings	The loading spectrum of the selected PC, with positive weights in blue and negative in red.	Peaks here are the Raman bands responsible for the variation along that PC — i.e. the chemistry behind the separation.
C — Score distribution	The spread of scores for the selected PC as a strip / box plot per group.	Shows how strongly each group expresses that component and how much they overlap.
D — Explained variance (scree)	The fraction of total variance captured by each PC.	The ‘elbow’ suggests how many components are meaningful; the rest is mostly noise.
E — Spatial score map	The selected PC's scores mapped back onto the sample, normalised to ±1 (positive vs negative), with a zero-score contour.	Turns an abstract component into a spatial chemical map of the sample.

Use the **Display PC** spinner (under *Spatial Score Map*) to choose which component drives panels B, C and E — changing it redraws the map. Click **Save PCA Figure** to export the whole panel.

Note The spatial score map (panel E) is available only when a single WDF map with a known pixel layout is loaded, because it needs the original X×Y geometry to reshape the scores. With XLSX input or multiple files, panels A–D still work.

11 Cluster Analysis

In brief: clustering automatically groups pixels with similar spectra into a chosen number of classes, producing a colour-coded class map and a representative spectrum per class.

Open from **Analysis › Cluster Analysis…**.

Control	Meaning
Number of clusters	How many groups to find (2–10).

Control	Meaning
Wavenumber min / max	Spectral window used for clustering.
K-means restarts	Number of random restarts for K-means (more = more stable).
Show cluster borders	Outline the boundaries between clusters on the map.
Offset mean spectra	Stack the per-cluster mean spectra vertically for clarity.

Click **Run Clustering**. Both K-means and agglomerative (hierarchical) methods are available. Save outputs with **Save Figure** or **Save Label Matrix (.csv)** — the latter exports the cluster index of every pixel.

12 MCR-ALS — Multivariate Curve Resolution

In brief: MCR-ALS decomposes the data into a small number of **pure component spectra** and their **abundance maps**, modelling each measured spectrum as a non-negative mixture — data ≈ abundances × pure spectra.

Open from **Analysis › MCR-ALS…**.

Control	Meaning
Non-negative abundances	Constrain abundances to be ≥ 0 (physically meaningful).
Non-negative spectra	Constrain recovered spectra to be ≥ 0.
Pixel bin (k×k)	Bin pixels before fitting to speed up the ALS iterations; full-resolution abundance maps are reconstructed afterwards.
float32	Use single precision (faster, lower memory; uses Apple M-series Accelerate where available).

Click **Run MCR-ALS**. From the results you can launch **Compare Components** (overlay the recovered spectra), **Identify Peaks** (annotate them against the band library), **Save Figure**, or **Save Pure Spectra (.csv)**.

13 N-FINDR — Endmember Extraction

In brief: N-FINDR finds the **purest** spectral signatures (endmembers) actually present in the data, then computes how much of each is present at every pixel.

Open from **Analysis › N-FINDR Endmembers…**.

Control	Meaning
Endmembers (p)	Number of pure signatures to extract.
Max N-FINDR iterations	Iteration cap for the search.
Wavenumber min / max	Spectral window used.

Click **Run N-FINDR**. Abundance maps are computed from the recovered endmembers by non-negative least squares (NNLS). Use **Compare Endmembers**, **Identify Peaks**, **Save Figure** or **Save Endmembers (.csv)**.

14 Peak Identification

New in v11. The Peak Identification window detects peaks in a selected spectrum and matches them against a Raman band library, listing each peak with its position (cm^{-1}), relative intensity and a candidate chemical assignment.

- Detected peaks are tabulated with columns: **cm^{-1}**, **Rel.I** (relative intensity) and **Assignment**.
- **Load custom CSV…** imports your own reference band list so the tool matches against your library.
- **Save Peak Table (.csv)** exports the identified peaks.

Peak Identification can also be launched directly from the MCR-ALS and N-FINDR results windows to annotate recovered component or endmember spectra.

15 Spectra Comparison

New in v11. The Compare / Superimpose window overlays two spectra so you can inspect differences directly. Choose **Spectrum A** and **Spectrum B**, and tick **Show difference (A – B)** to plot their difference trace. It also compares clustering components and extracted endmembers, and the comparison figure can be exported as PNG or PDF.

16 Spectral Tools

Open from **Analysis › Spectral Tools…**. These four utilities modify the loaded map **in place**, so save your work first if you want to keep the original.

Tool	What it does
1 · Spectral Resampling	Resample all spectra onto a new equidistant wavenumber grid — set start, end (in cm^{-1}) and number of points, then **Apply Resampling**.
2 · Spatial Crop	Crop the map to a pixel bounding box. **Set from Map Size** fills the current extent; adjust the bounds and **Apply Crop**.
3 · Map Rotation	Rotate the map by 0 / 90 / 180 / 270 degrees, then **Apply Rotation**.
4 · Background Subtraction	Subtract an optical substrate / background reference, with a subtraction scale from 0 to 2, then **Apply Subtraction**.

Note Because these operations are applied in place, the change persists for the rest of the session. Reload the original file to undo them.

17 3D Confocal Volume Viewer

In brief: the 3D viewer turns a depth-resolved (confocal) Raman dataset into an interactive three-dimensional rendering, so you can see how chemical components are distributed through the **volume** of a sample rather than just across its surface. It is designed to reproduce the kinds of renders produced by HORIBA LabSpec, and works either from a true Z-stack or — for a quick look — from a single 2D map.

Open from **Analysis › 3D Volume Viewer…**. All controls are on the left; the render appears on the right after you press **Render 3D View**.

Step 1 — Provide the volume data

Set the **Z step (µm)** first (the physical spacing between depth planes; default 1.0), then click **Load Z-stack WDF files…** and select *all* the slice files at once — they are sorted and stacked, and each is assigned a depth of its index × the Z step. `.npy` arrays are also accepted. If you load nothing, the viewer falls back to the current single 2D map and synthesises a pseudo-depth axis from Band B, which is enough to preview the render modes.

Step 2 — Choose a render mode

Render mode	Description
Volume Scatter	Voxels above the threshold are drawn as a coloured 3D point cloud. Band A drives the colour map; Band B is added as a green overlay. Good for seeing where a component concentrates in space.
Orthogonal Slices	Interactive XY / XZ / YZ cross-section planes through the volume; set their positions with the Slice index controls (below).
3D Surface	Renders the intensity map as a 3D height surface, with an optional second-band overlay — useful for topography-style views.
Multi-band RGB	Band A → Red, Band B → Green, Band C → Blue, combining three components into one false-colour volume (HORIBA-style polymer / geological renders). Requires Band C to be enabled.

Step 3 — Define the spectral bands

Each band is an integration window set by a **Low** and **High** wavenumber (cm^{-1}); Bands A and B also take a text **Label** used in the legend. Band A is the primary (red) signal, Band B the secondary (green), and Band C (blue) is optional — tick **Enable Band C** to use it, which is required for Multi-band RGB mode. The default windows are pre-filled around the centre of your spectral range.

Step 4 — Adjust the display

Control	Default	Effect
Threshold (0–1)	0.15	Voxels below this normalised intensity are hidden — raise it to show only the strongest signal, lower it to reveal faint structure.
Voxel alpha	0.55	Transparency of the rendered voxels; lower values let you see through the cloud.
Scatter pt size	12	Size of points in Volume Scatter mode.
Z scale factor	1.0	Stretches or compresses the depth axis for clarity (exaggerate thin stacks, compress deep ones).
Pre-smooth σ (px)	0.8	Gaussian smoothing applied before rendering to reduce noise; 0 disables it.
Colourmap	hot	Colour scheme for intensity (Volume Scatter / Surface modes).

Step 5 — Lighting, orientation & slices

Control	Default	Effect
Dark background	On	Black backdrop (better contrast for glowing renders) vs white.
Show axis labels	On	Toggle the X / Y / Z axis annotations.
Show bounding box	On	Toggle the wireframe box around the volume.
Elevation (°)	25	Vertical viewing angle of the 3D camera.

Control	Default	Effect
Azimuth (°)	–60	Horizontal rotation of the 3D camera.
Z / Y / X slice index	0	Positions of the cross-section planes in **Orthogonal Slices** mode.

Step 6 — Render and export

Click **Render 3D View** to build and draw the volume (large stacks take a moment — a 'Building volume…' message shows progress). Adjust any control and re-render as needed. Click **Export PNG / PDF** to save the current view at 250 dpi.

Note Rendering is the most memory-intensive feature. If it is slow, raise the threshold, increase pre-smoothing, or reduce the volume size before rendering.

18 Exporting Results

Most windows offer their own save buttons. The common outputs are:

Output	Where	Format
Current map	File › Save Map (Ctrl+M)	image
Current spectrum	File › Save Spectrum (Ctrl+S)	CSV / image
Analysis figures	Save Figure in each analysis window	PNG / PDF
Cluster labels	Cluster Analysis › Save Label Matrix	CSV
Pure spectra / endmembers	MCR-ALS / N-FINDR save buttons	CSV
Peak table	Peak Identification › Save Peak Table	CSV
3D render	3D Viewer › Export PNG / PDF	PNG / PDF (250 dpi)

19 Keyboard Shortcuts

Shortcut	Action
Ctrl+O	Open a WDF map file
Ctrl+M	Save the current map
Ctrl+S	Save the current spectrum
Right-click on map	Add the clicked spectrum to the comparison overlay
Left-click on map	Select a pixel / show its spectrum

20 Troubleshooting & FAQ

WDF loading is disabled / greyed out

The **renishawWiRE** package is not installed. Install it with `pip install renishawWiRE` and restart the program.

Baseline options are missing

The **pybaselines** package is not installed. Install it with `pip install pybaselines`.

My preprocessing changes had no effect

Preprocessing settings apply to the **next** file loaded. Reload the data after changing settings, and check the Processing Log to confirm what was applied.

An analysis is very slow

Large maps are demanding. Use a wavenumber window to limit the spectral range, sub-sample spectra (PCA), enable pixel binning (MCR-ALS), or use float32 where offered.

I changed a map with Spectral Tools and want the original back

Spectral Tools edit the map in place. Reload the original `.wdf` file to restore it.

21 Glossary

Term	Meaning
Hyperspectral map	A dataset with a full spectrum recorded at every pixel.
Wavenumber (cm^{-1})	The x-axis of a Raman spectrum; position of spectral features.
Baseline correction	Removal of broad background (e.g. fluorescence) under the peaks.
ROI	Region of Interest — a sub-area selected for analysis.
PCA	Principal Component Analysis; finds main axes of spectral variation.
Clustering	Grouping pixels with similar spectra into classes.
MCR-ALS	Multivariate Curve Resolution — Alternating Least Squares; unmixes data into pure spectra and abundance maps.
Endmember	A pure spectral signature present in the data (extracted by N-FINDR).
NNLS	Non-Negative Least Squares; computes non-negative abundances.
LUT	Look-Up Table; the mapping from intensity values to display colours.

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